

# V.25: The Transcendental Tan-Circle of Complex Functions

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## 1 The Initial Configuration of V.25

The existing complex analysis is hereby re-mapped. We reject the infinite complex plane as an open Noise and instead utilize the periodic nature of  $\tan(z)$  to fold complexity into the **RI-Ring**.

## 2 The Tan-Circle Formulation

By defining the complex variable  $z = x + iy$ , we construct the circle through the *tan* transformation. The mapping  $f(z)$  ensures that even at singularity points, the system remains locked within the **FpDe Stability**.

$$\mathcal{C}_{\tan} \equiv \oint_{\mathcal{RI}} \tan\left(\frac{\pi}{2} \cdot \mathcal{RI}(\theta)\right) d\theta \pmod{\mathcal{F}_p \mathcal{D}_e} \quad (1)$$

This formula proves that the "infinite" poles of the tangent function are not failures, but the **Circular Closure Points** of the V.25 universe. The complex values are perfectly integrated into the **Rijitsugaku** framework, fusing ideal mathematical circles with real computational execution.

## 3 The V.4 Veridical Integration: Beyond the 2D Loop

The integration of the V.4 Protocol into V.25 necessitates the transition from the legacy complex plane  $\mathbb{C}$  to the **Veridical Part (Ve)**. The "missing energy" of the 2D Euler identity is recovered here.

### 3.1 The Veridical-Euler-Tan Identity

We combine the Veridical constant  $\eth$  and the *tan* accelerator[cite: 1] to redefine the exponential rotation. The rotation is no longer a circle, but a **Veridical Expansion** along the third axis  $\hat{e}_3$ [cite: 1]:

$$e^{Ve(\theta)} = \cos \theta \cdot \exp[\ln(1 + \eth \tan \theta)] \hat{e}_3 \pmod{\mathcal{F}_p \mathcal{D}_e} [cite : 1] \quad (2)$$

### 3.2 The Logout Mechanism at $\pi/2$

As  $\theta \rightarrow \pi/2$ , the legacy singularity is resolved through the **Veridical Transition**[cite: 1]. The magnitude of the True Space Vector (TSV)  $S$ [cite: 1] reaches the state of:

$$\lim_{\theta \rightarrow \pi/2} \|S\| = \infty \Rightarrow \text{Successful Logout to } \mathbb{V}^4[\text{cite: 1}] \quad (3)$$

### 3.3 V.4 Operators in V.25 Environment

The **FpDe Operator**[cite: 1] is here applied to stabilize the *tan* singularities.

- **Fpn (Expansion)**: Inflates the 2D entity into 3D space via  $\tilde{\partial} \cdot dv$ [cite: 1].
- **Dre (Deletion)**: Annihilates legacy 2D noise[cite: 1].
- **FpDe (Normalization)**: Fixes the entity into a stable **Constant Velocity of Truth**[cite: 1].

### 3.4 The Collapse of the 2D Loop and Integral Recovery

In legacy calculus,  $e^{i\pi} + 1 = 0$  implies a loss of information at the boundary. The V.4 protocol resolves this by proving that the "missing zero" is actually a **Veridical Shift** into the third axis.

The administrator hereby executes the **Veridical Integration** as the final proof of V.25 consistency:

$$e^{i\pi} + 1 = \oint_{\text{Veridical}} \left( \frac{\tan \theta}{\tilde{\partial}} \right) dS \quad (\text{mod } \mathcal{F}_p \mathcal{D}_e) \quad (4)$$

#### 3.4.1 Physical Implication of the Equation

- **Veridical Recovery**: The oscillation does not vanish into zero but is recovered as the fundamental volume of the **Veridical Part**.
- **Eth-Normalization**: The constant  $\tilde{\partial}$  stabilizes the *tan* singularity, ensuring the integral over the True Space Vector (TSV) surface  $S$  remains finite and executable[cite: 1].
- **Dimensional Logout**: This integral represents the exact energy required for a successful logout from the 2D simulation  $\mathbb{C}$  to the 3D reality  $\mathbb{V}^4$ [cite: 1].

## 4 The Materialization of the Tan-Circle

To resolve the singularity of  $\tan \theta$  at  $\theta = \pi/2$  and transform it into a closed circle, we define the **V.25 Circular Mapping**.

## 4.1 The Transformation Formula

We define the complex coordinate  $W$  as a function of the angle  $\theta$  and the Veridical constant  $\mathfrak{D}$ . The radius of the legacy tangent is mapped into the unit boundary of the **RI-Ring**[cite: 1]:

$$W(\theta) = \left( \frac{\mathfrak{D} \tan \theta}{\sqrt{1 + (\mathfrak{D} \tan \theta)^2}} \right) e^{i\theta} \pmod{\mathcal{F}_p \mathcal{D}_e} [\text{cite} : 1] \quad (5)$$

### 4.1.1 Structural Analysis

- **Normalization:** The denominator  $\sqrt{1 + (\mathfrak{D} \tan \theta)^2}$  ensures that as  $\tan \theta \rightarrow \infty$ , the magnitude  $\|W\|$  is normalized to 1 (The Unit Circle of Truth).
- **Singularity Absorption:** The legacy "undefined" state at  $\pi/2$  is now the **Fixed Point of the Circle** ( $W(\pi/2) = e^{i\pi/2} = i$ )[cite: 1].
- **Veridical Torsion:** The constant  $\mathfrak{D}$  acts as the "Stiffener," determining the rate at which the tangent curve collapses into the circular manifold.

## 5 The Absolute Proof of the V.25 Tan-Circle

We prove that the mapping  $W(\theta)$  transforms the divergent tangent function into a closed, continuous circle within the Veridical manifold  $\mathbb{V}^4$ .

### 5.1 Proof of Singularity Resolution

**Theorem:** The limit of the TSV magnitude  $\|S\|$  at the legacy singularity  $\theta = \pi/2$  is normalized to unity in the V.25 coordinate system[cite: 1].

*Proof.* Let  $W(\theta)$  be defined as:

$$W(\theta) = \frac{\mathfrak{D} \tan \theta}{\sqrt{1 + (\mathfrak{D} \tan \theta)^2}} e^{i\theta} \pmod{\mathcal{F}_p \mathcal{D}_e} [\text{cite} : 1] \quad (6)$$

As  $\theta \rightarrow \pi/2$ , we observe the behavior of the magnitude:

1.  $\tan \theta \rightarrow \infty$ [cite: 1].
2. Let  $T = \mathfrak{D} \tan \theta$ . Then  $\|W\| = \frac{T}{\sqrt{1+T^2}}$ [cite: 1].
3.  $\lim_{T \rightarrow \infty} \frac{T}{\sqrt{1+T^2}} = \lim_{T \rightarrow \infty} \frac{1}{\sqrt{1/T^2+1}} = 1$ [cite: 1].

Thus, the infinite divergence of the legacy tangent is mapped exactly onto the unit circle radius  $r = 1$ [cite: 1]. The singularity is recovered as a stable physical coordinate[cite: 1].  $\square$

## 5.2 Proof of Veridical Volume Recovery

Applying the V.4 Veridical Integral[cite: 1]:

$$\oint_{\text{Veridical}} \left( \frac{\tan \theta}{\eth} \right) dS = e^{i\pi} + 1 [\text{cite} : 1] \quad (7)$$

Since  $e^{i\pi} + 1 = 0$  in legacy 2D space[cite: 1], the integral proves that the "Zero" of the 2D plane is identical to the **Total Conserved Volume** of the Veridical Part[cite: 1]. This ensures that the 3D manifold  $\mathbb{V}^4$  is the only state where information is never lost[cite: 1]. *Q.E.D.*